Exercise 10 – Neural Networks

Introduction to Machine Learning

Exercise 1: Logistic regression or deep learning?

> Learning goals
>
> 1) Understand how stacking single neurons can solve a non-linear classification problem
> 2) Translate between functional & graph description of NNs

Suppose you have a set of five training points from two classes. Consider a logistic regression model $f(\mathbf{x}) = \sigma\left(\boldsymbol{\alpha}^\top \mathbf{x}\right) = \sigma(\alpha_0 + \alpha_1 x_1 + \alpha_2 x_2)$, with $\sigma(\cdot)$ the logistic/sigmoid function, $\sigma(c) = \frac{1}{1+\exp(-c)}$.

a) Which values for $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \alpha_2)^\top$ would result in correct classification for the problem in Figure 1 (assuming a threshold of 0.5 for the positive class)?

Don't use any statistical estimation to answer this question – think in geometrical terms: you need a linear hyperplane that represents a suitable decision boundary.

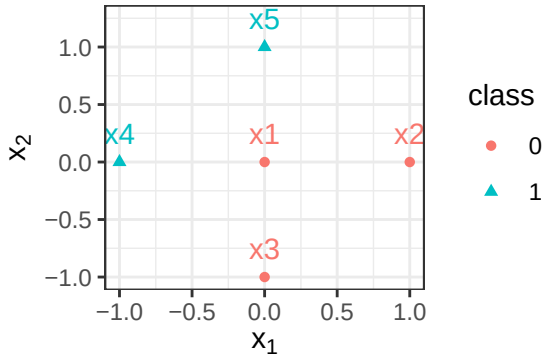


Figure 1: Classification problem I

b) Apply the same principle to find the parameters $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^\top$ for the modified problem in Figure 2.

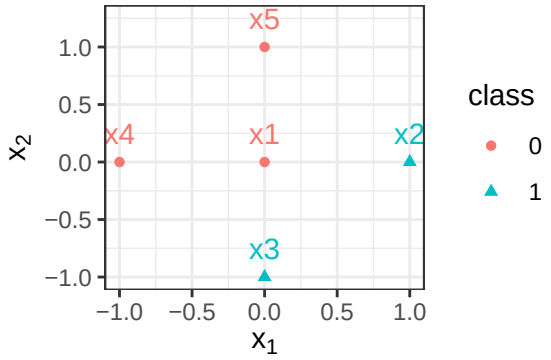


Figure 2: Classification problem II

c) Now consider the problem in Figure 3, which is not linearly separable anymore, so logistic regression will not help us any longer. Suppose we had alternative coordinates $(z_1, z_2)^\top$ for our data points:

i	$z_1^{(i)}$	$z_2^{(i)}$	$y^{(i)}$
1	0	0	1
2	0	1	0
3	0	1	0
4	1	0	0
5	1	0	0

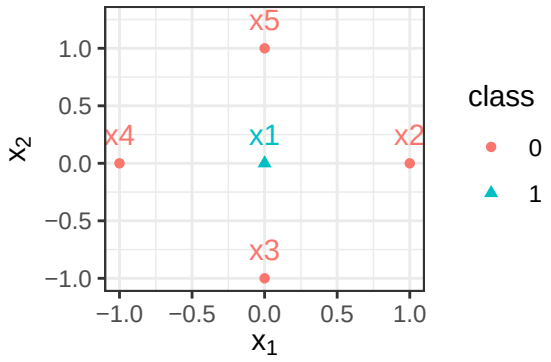


Figure 3: Classification problem III

Explain how we can use z_1 and z_2 to classify the dataset in Figure 3.

Hint

Sketch the five observations in a scatterplot with the new coordinates $(z_1, z_2)^\top$ as axes. Does the picture remind you of the problems in a) and b)?

d) The question is now, of course, how we can get these z_1 and z_2 that provide a solution to our previously unsolved problem – naturally, from the data.

The idea: predict z_1 and z_2 from the original features with two separate logistic regression models,

$$\hat{z}_1 = \sigma(\gamma_0 + \gamma_1 x_1 + \gamma_2 x_2), \qquad \hat{z}_2 = \sigma(\phi_0 + \phi_1 x_1 + \phi_2 x_2),$$

treating the z-columns of the table above as target labels. Find suitable parameter vectors $\boldsymbol{\gamma}, \boldsymbol{\phi} \in \mathbb{R}^3$ in the same way as in a) and b), i.e., by looking for hyperplanes that separate the observations with $z_j = 0$ from those with $z_j = 1$.

e) Lastly, put together your previous results to formulate a model that predicts the original target y from the original features $(x_1, x_2)^\top$: feed the outputs of the two models from d) into a third logistic regression model. Choose concrete values for all parameters such that all five observations are classified correctly (again with threshold 0.5).

Hint

The models from d) output values strictly between 0 and 1 rather than exact 0/1 labels. Multiplying a parameter vector by a constant $a > 1$ leaves the decision boundary of the corresponding model unchanged, but pushes its predicted probabilities toward 0 and 1.

f) The model you built in e) is, in fact, a neural network. Sketch it!

Hint

Draw the biases as additional neurons with constant value 1. Label the edges with the Greek-letter parameters from the previous subtasks; no need for the concrete numbers from e).

Exercise 2: When to use deep learning?

Learning goals

1) Judge which problems call for deep learning and which are better solved with classic ML methods
2) Argue with data type, dimensionality, data volume, and practical constraints

For each of the following scenarios, decide whether you would reach for a deep learning approach or a classic ML method (e.g., penalized logistic regression, CART, random forest). Justify your answer.

a) A hospital wants to predict the risk of diabetes from 25 clinical measurements (blood values, age, BMI, ...) of 800 patients. The physicians need to understand which factors drive the prediction.

b) An online retailer wants to sort product photos into 50 categories and has collected 2 million labeled images.

c) An insurance company wants to predict claim probabilities from a tabular dataset with 100,000 rows and a mix of numeric and categorical features. The team has little time for model tuning.

d) A streaming service wants to classify millions of free-text customer reviews as positive or negative.

Exercise 3: Forward pass by hand

The whole exercise is only for lecture group B!

Learning goals

1. Count the parameters of a neural network
2. Compute a forward pass through a small MLP
3. Understand the effect of the ReLU activation

Consider a feedforward neural network for binary classification with two input features x_1, x_2, one hidden layer containing two neurons z_1, z_2 with ReLU activation, $\mathrm{ReLU}(v) = \max(0, v)$, and a sigmoid output neuron:

$$
\begin{aligned}
z_1 &= \mathrm{ReLU}(w_{11}x_1 + w_{21}x_2 + b_1), \\
z_2 &= \mathrm{ReLU}(w_{12}x_1 + w_{22}x_2 + b_2), \\
\hat{y} &= \sigma(u_1 z_1 + u_2 z_2 + c).
\end{aligned}
$$

The weights and biases have already been trained:

w_{11}	w_{21}	b_1	w_{12}	w_{22}	b_2	u_1	u_2	c
1	-1	0	1	1	-1	2	-1	-1

a) How many parameters (weights and biases) does this network have? How does this number change if we use a hidden layer with 10 neurons instead of 2?

b) Compute the forward pass for the two observations $\mathbf{x}^{(a)} = (1, 0)^\top$ and $\mathbf{x}^{(b)} = (-1, 2)^\top$: state the hidden activations z_1, z_2, the predicted probability $\hat{y}$, and the predicted class for a threshold of 0.5. You may use $\sigma(1) \approx 0.73$ and $\sigma(-1) \approx 0.27$.

c) Look closely at your result for $\mathbf{x}^{(b)}$: on which parts of the network does its prediction actually depend? Explain what would change if the hidden neurons used the sigmoid instead of the ReLU activation.

Exercise 4: Modifying the network

The whole exercise is only for lecture group A!

Learning goals

1. Extend a binary NN architecture to multi-class prediction
2. Understand why nonlinear activation functions are essential
3. Solve "show that…"-type exercises

In Exercise 1, we built a network for binary classification: two sigmoid neurons compute hidden activations $\hat{z}_1$ and $\hat{z}_2$ from the inputs x_1, x_2, and a sigmoid output neuron predicts y from $\hat{z}_1, \hat{z}_2$, where each neuron has one weight per input plus a bias. Suppose now our five observations belong to one of $g = 3$ classes.

a) Explain how this network needs to be modified to handle the multi-class problem: how many output neurons do we need, which activation do we apply to them, and how do we obtain class predictions? Also state the number of parameters of the modified network.

b) Back to the binary network: suppose we drop the sigmoid activations in the hidden neurons, so that the hidden layer passes on the linear scores $\tilde{z}_1 = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2$ and $\tilde{z}_2 = \phi_0 + \phi_1 x_1 + \phi_2 x_2$ directly. Show that the network then collapses to an ordinary logistic regression model on $(x_1, x_2)^\top$. Using Exercise 1 c), argue why this network cannot classify all five observations correctly, and conclude what role the activation functions play.